

Avalon Reference Design System™

Avnet Design Services has created the Avalon Reference Design System - a compatible set of system modules used by designers to create a custom development environment targeted for their specific application. The Avalon Reference Design System provides a robust set of system design modules at a number of system levels by attracting industry leaders - from hardware components, IP cores, firmware drivers, applications code, CAD development software, test and analysis tools, to training and hands-on workshops, qualified design services and electronics manufacturing services providers. Avnet Design Services has attracted the participation of industry leading manufacturers and service providers for each of the required system design elements of the Avalon Reference Design System. Avalon participants contribute their unique expertise for their specific area, giving the Avalon environment both the depth and breadth of expertise required to construct best-in-class solutions.

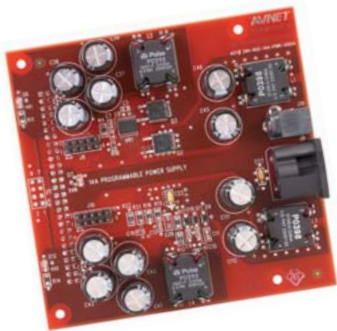
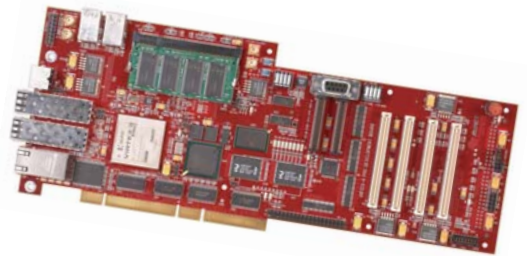
Xilinx Virtex-II Pro™ Development Kit

The Virtex-II Pro Development Kit from Avnet Design Services offers system level development on a highly advanced hardware platform. The Xilinx Virtex-II Pro family, based on the highly successful Virtex-II architecture, provides a unique environment for developing high-performance microprocessor and I/O intensive applications. The embedded 32-bit IBM PowerPC™ RISC processor delivers over 420 Dhrystone MIPS at 300 MHz, with a high-bandwidth interface to on-chip programmable logic. Additionally, Rocket I/O™ 3.125 Gbps transceivers support emerging connectivity standards. Time to market is greatly accelerated for implementation of advanced peripherals, IP and high-speed protocols such as 10 Gig Ethernet, Fibre Channel, OC-192 and InfiniBand.

ADS-XLX-V2PRO-DEVP7-5 \$1,995.00 USD resale - Populated with XC2VP7, - 5 speed grade

ADS-XLX-V2PRO-DEVP7-6 \$2,495.00 USD resale - Populated with XC2VP7, - 6 speed grade

ADS-XLX-V2PRO-DEVP20-5 \$2,995.00 USD - Populated with XC2VP20, - 5 speed grade



National Semiconductor Dual High-Current Power Supply Kit

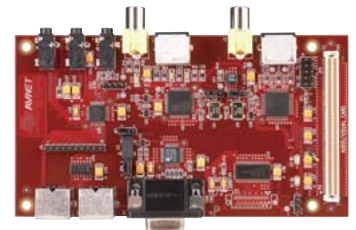
The Avnet designed National Semiconductor 28A DC-DC power supply board is a dual programmable Synchronous Buck Regulator. The design utilizes two National Semiconductor LM2636 Synchronous controllers. Each controller is capable of supplying 14 amps with variable output voltages ranging from +1.3VDC to +3.3VDC. The board also has a +5VDC feed-through from the input source that can be used to supply a third voltage rail to the target board. A status signal indicating voltage is good on either output rail is also available at the target as well as an enable pin to shut both supplies off.

ADS-NSC-XP \$250.00 USD resale

Audio/Video Module

The Audio/Video Module is an expansion daughtercard for use with other Avalon Reference Design System™ compliant products. It offers a platform to develop and test products that require audio and/or video. Key components that can be accessed through the AvBus connector include Philips' UCB1400 Audio Codec, Philips' SAA7111A Enhanced Video Input Processor, Philips' SAA7121H Digital Video Encoder, Analog Devices' ADV7123 Triple Video DAC, and National Semiconductor's DS90C383 LVDS transceiver.

ADS-AV-DAU \$300.00 USD resale



Motorola MCF5282 Evaluation Kit

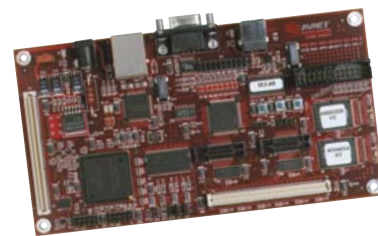
The Motorola ColdFire® Evaluation Board provides a reference platform for engineers who need to develop a variety of embedded processing applications requiring networking connectivity. The board hosts all the necessary hardware and firmware needed to implement a networked interface using the Motorola MCF5282 processor. By including all the necessary PHY devices, memory and external expansion capability, the designer can implement any bridging application that requires 10/100 Ethernet, UARTs, CAN interface, QSPI, analog inputs and/or memory-mapped peripherals.

ADS-MOT-5282-EVL \$250.00 USD resale

Motorola MPC857T Evaluation Kit

Enhance engineering productivity and accelerate time to market with the MPC857T Evaluation Kit from Avnet Design Services. The MPC857T delivers an integrated platform to develop and test designs targeted to the highly advanced Motorola MPC857T Integrated Communications Processor. The MPC857T PowerQUICC™ processor offers a versatile one-chip integrated microprocessor and peripheral combination that can be used in a variety of controller applications. It particularly excels in both communication and networking systems demonstrating the enhanced benefits of the leading edge network controller solution. A variety of memory technologies are implemented to allow full utilization of the MPC857T controller functions. The kit features 10/100 Ethernet, USB 1.1 Type B, and RS-232 console interfaces as well as AvBus connectors for host system and daughter card expansion. A complete software development suite is included with the kit for maximum design flexibility.

ADS-MOT-857-EVL \$499.00 USD resale



Xilinx and Cypress USB 2.0 to SCSI System Solution Kit

Avnet Design Services, Xilinx, Cypress and Mentor Graphics have created the USB 2.0 to SCSI System Solution Kit using the low-cost Xilinx Spartan™-IIE family of FPGAs and Cypress' CY7C68000 USB PHY. USB is the dominant peripheral interface for PCs, offering auto configuration, hot pluggability, low-cost and high reliability.

This kit is a complete application solution and provides the designer with a Windows ready USB 2.0 to SCSI demonstration design. The kit contains complete design documentation, schematics, layout and a license to run a demonstration bit file programmed into the on-board PROM. Additionally, customers can purchase a license for the USB 2.0 IP Core and SCSI IP Core used in the Xilinx Spartan-IIE device for production.

ADS-S2E-US2-SOL \$499.00 USD resale

Xilinx Virtex™-II Development Kit *MicroBlaze*

The Virtex-II Development Kit provides a complete hardware environment for designers to accelerate their time to market. The kit delivers a stable platform to develop and test designs targeted to the highly advanced Xilinx Virtex-II FPGA family. The installed FPGA device offers up to a 6 Million system gate prototyping environment to effectively demonstrate the enhanced benefits of the leading edge Xilinx FPGA solution. Embedded multipliers, advanced digital clock management, built-in impedance matching, IP immersion, and other exciting features can be implemented with advanced Xilinx design tools. Demonstration VHDL code is included with the kit to exercise standard peripherals on the evaluation board for a quick start to device familiarization.

ADS-XLX-V2-DEV1500 \$1,000.00 USD resale – Populated with an XC2V1500 device

ADS-XLX-MB-DEV1500 \$1,400.00 USD resale – Bundled with Communications/Memory Module and MicroBlaze Core License

ADS-XLX-V2-DEV4000 \$2,500.00 USD resale – Populated with an XC2V4000 device

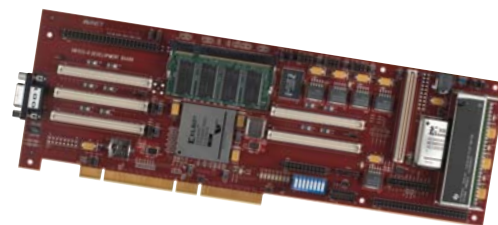
ADS-XLX-MB-DEV4000 \$2,900.00 USD resale – Bundled with Communications/Memory Module and MicroBlaze Core License

ADS-XLX-V2-DEV4000XP \$3,000.00 USD resale – XC2V4000 & high-current power supply

ADS-V2-MB-DEV4000XP \$3,400.00 USD resale – Bundled with Communications/Memory Module and MicroBlaze Core License & high-current power supply

ADS-XLX-V2-DEV6000XP \$6,000.00 USD resale – XC2V6000 & high-current power supply

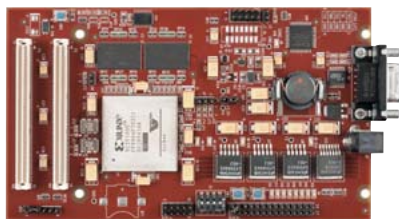
ADS-V2-MB-DEV6000XP \$6,400.00 USD resale – Bundled with Communications/Memory Module and MicroBlaze Core License & high-current power supply



Denali RLD RAM IP Core Demonstration Kit

The Denali RLD RAM IP Core Demonstration Kit is equipped with I/O cell circuitry used for compensation to meet timing requirements, such as: programmable read clock delay and programmable write clock delay. The IP Core has two separate read clocks, each driven by a DQS signal from a memory device. DDR I/O cells in conjunction with the on board DCMs of the FPGA maximize the data valid window to reliably capture and send data to and from the RLD RAM devices. ECC and BIST circuitry are also ready to be implemented if necessary.

ADS-DLI-RLDRAM Contact your local Avnet representative - Denali core bundled with RLD RAM daughtercard



Communications/Memory Module *MicroBlaze*

The Communications/Memory module is an expansion daughtercard for use with the Avalon Reference Design System™. The daughtercard interfaces via AvBus connectors and provides general purpose resources to complement Avalon based modules. These include 10/100/1000Base T Ethernet, 16 MB FLASH, 64 MB SDRAM, 1 MB SRAM, IrDA, PC Card and USB 2.0. The daughtercard provides all necessary resources for implementation of Xilinx MicroBlaze™ core designs.

ADS-EMU-DAU \$250.00 USD resale

ADS-XLX-MB-DAU \$499.00 USD resale – Bundled with MicroBlaze Core License



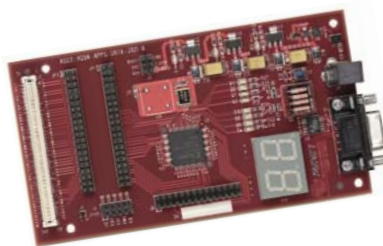
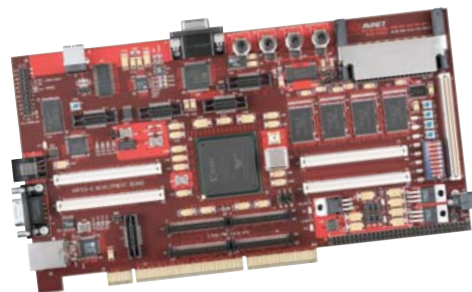
Xilinx Virtex™-E Development Kit *MicroBlaze™*

The Xilinx Virtex-E family of FPGAs ranges in size from 70,000 to 4,000,000 system gates and delivers low-power 130 MHz system performance. The Virtex-E family combines a wide variety of programmable system features, including a rich hierarchy of fast and flexible interconnect resources, sophisticated SelectRAM+™ memory and SelectIO+™ interface technology.

The Virtex-E Development Platform provides a complete hardware environment for designers to accelerate time to market. Unlike other boards that offer limited I/O ports, no memory and little expansion capability, the Virtex-E Development Platform offers a majority of functions usually required by system designers. The Virtex-E Development Platform offers the ability to add custom hardware via PCI bus, a PC card connector, PCI mezzanine connector, and AvBus™ modular hardware expansion connectors. Additional features include an XCV1000E-6FG1156 FPGA, 64 MB SDRAM, 32 MB Flash, RS-232 port, CAN PHY interface, USB, video decoder, triple video DAC, and stereo DAC.

ADS-XLX-VE-DEV \$1,495.00 USD resale

ADS-VE-MB-DEV \$1,650.00 USD resale - Bundled with MicroBlaze Core License



Xilinx CoolRunner™-II Evaluation Kit

The Xilinx CoolRunner-II Evaluation Kit from Avnet Design Services speeds time to market for designers, offering a platform to develop and test products that are targeted to the Xilinx CoolRunner-II CPLD family. The CoolRunner-II offers high performance and ultra low power consumption in a single device without compromises. With less than 100 µA of standby current, CoolRunner-II is a perfect solution for any new system. The kit features a serial A/D converter which displays the CPLD current consumption, RS-232, user interface, and AvBus expansion capability.

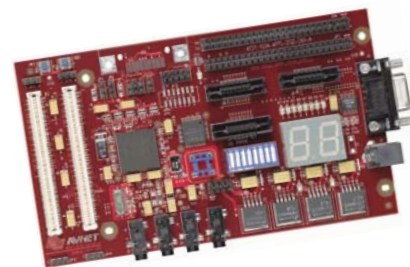
ADS-XLX-CR2-EVL \$149.00 USD resale

Xilinx Spartan™-IIE Evaluation Kit *MicroBlaze™*

The Xilinx Spartan-IIE Evaluation Kit from Avnet Design Services speeds time to market for designers, offering a platform to develop and test products that are targeted to the Xilinx Spartan-IIE FPGA family. Spartan-IIE devices represent the first FPGA specifically optimized for the low-cost, rapidly changing consumer digital convergence market. With its advanced silicon, industry-leading software, and extensive IP library, the Spartan-IIE solution is the perfect product for addressing the changing design paradigm triggered by the digital convergence of consumer products. The Spartan-IIE Evaluation Kit also assists newcomers with FPGA implementation and VHDL design by providing a stable environment in which to work, complete with tested demonstration designs. A Xilinx XC2S200E device, capable of implementing 200,000 system gates, is installed as the target for designs. The Xilinx Spartan-IIE Evaluation Kit is a re-programmable development and prototyping hardware platform that will shorten and simplify design cycles.

ADS-XLX-SP2E-EVL \$249.00 USD resale

ADS-SP2E-MB-EVL \$650.00 USD resale – Bundled with Communications/Memory Module and MicroBlaze Core License



Xilinx PMC IRL™ Reference Design Kit

The 'Avnet Design Services' Xilinx PCI Mezzanine Card (PMC) Internet Reconfigurable Logic (IRL) Reference Design Kit provides the user with a complete development environment on which PAVE Framework and other IRL applications can be developed. Designers can begin development immediately on a flexible hardware platform using a working software environment - dramatically cutting time to market for IRL based applications. Included in the kit is a powerful Virtex-II based FPGA board that includes sufficient memory and interfaces to cover most applications requirements, the PAVE Framework software toolset which provides an easy to use set of standard IRL software modules and interfaces, documentation, schematics and a bill of materials so all or part of the design can be easily included in a final design.

ADS-XLX-PMC-IRL \$2,495.00 USD resale



Motorola 68HC908GP32 emWare® Enabled Development Kit

The Motorola 68HC908GP32 emWare Enabled Development Kit combines all the features and functionality of Motorola's 68HC908GP32 8-bit FLASH microcontroller with emWare's EMIT® device networking software to provide a powerful, low-cost development solution. emWare's EMIT device networking technology is a complete software infrastructure that allows system designers to produce intelligent devices that can be managed and controlled remotely over the Internet or any other network, while adding as little as 1KB of code.

Available exclusively from Avnet Design Services, the Motorola 68HC908GP32 emWare Enabled Development Kit also doubles as a microcontroller target system, significantly reducing designer development requirements and product time-to-market.

ADS-MOTGP32-EMW \$249.95 USD resale



Cypress Delta39K™ Evaluation Kit

The Delta39K CPLDs are the first to offer large blocks of single-port, dual-port and FIFO communications RAM and are supported by the industry leading Cypress Warp™ design tools. The evaluation board features an analog interface circuit, an RS-232 port, switches, LED displays and expansion capability.

ADS-CYP-39K-EVL \$550.00 USD resale

Intel StrongARM PCI Development Kit

For designers considering the Intel 32-bit StrongARM SA1110 processor for the next embedded design, this kit can save you time and effort. In a compact PCI form factor, this kit is packed with commonly used interfaces and plenty of memory to get the development process going quickly.

Hardware features include 64MB SDRAM, 48MB FLASH, PCI interface, 10/100 Ethernet port, USB port, RS-232 port and support for PCMCIA with the creation of an adapter board. A Xilinx Spartan™-II FPGA implements a glueless interface between the SA1110 processor bus and the PCI bus. Included in the cost of the kit are the development board, Orcad schematics, Intel StrongARM Bus Interface VHDL source code, Xilinx PCI core single use license agreement, 16 hours of FPGA design support and complete documentation.

ADS-ISTA-PCI-KIT \$8,995.00 USD resale

Motorola M68HC08 Family Evaluation Kit, 908JL3 Version

The 68HC908JL3 is a low power, static, 8-bit microcontroller with 4KB of FLASH, 128 bytes of RAM, a 12-channel, 8-bit A/D, a 2-channel 16-bit timer, and 23 I/O pins. Rated for operation at 3V and 5V, the 908JL3 is in-system programmable.

The board in the JL3 kit features a prototype area, RS-232 port, two 7-segment LEDs, temperature sensor, potentiometer for A/D measurements, and a pressure sensor.

The included demo software routines use the 2-digit LED display to show the fractional value of resistance of the potentiometer, the temperature, or the barometric pressure.

ADS-MOT-JL3-EVL \$89.95 USD resale

Motorola 68HC908GP32 Evaluation Kit

Motorola's 68HC908GP32 is a member of the popular M68HC08 family of microcontrollers. The GP32 features 32KB of FLASH memory, 512 bytes RAM, SPI module, SCI module and an 8-channel, 8-bit successive approximation analog-to-digital converter.

An LCD display, temperature sensor, piezoelectric speaker and RS-232 port add a high degree of functionality to this low cost kit. With the included assembler/debugger and power supply, designers have everything they need to start working with the 68HC908GP32 today.

ADS-MOT908GP32 \$149.95 USD resale

AvBus Breakout Module

This daughtercard provides a convenient method of interfacing external signals to standard AvBus connectors on development and evaluation boards. The board features two male and two female AvBus connectors with breakout to six MICTOR connectors and four 50-pin (0.1") headers. Power indicator LEDs are provided for +5 and +3.3 VDC.

ADS-AVBUS-DAU \$125.00 USD resale

Avnet Parallel Cable III

The Avnet Parallel Cable III allows configuration of Xilinx FPGAs, PROMs and CPLDs quickly and easily. The cable can be used in conjunction with Xilinx ISE development software or WebPack device programmer and supports a parallel port connection to a PC. The Parallel Cable III is a very cost effective solution to enable JTAG and Xilinx serial configurations. The cable comes complete with two sets of "fly wires" which connect the Parallel Cable to a circuit board.

HW-JTAG-PC \$95.00 USD resale

Xilinx Spartan™-II Evaluation Kit

This kit includes a Xilinx XC2S150-5PQ208 FPGA, Xilinx SPROM, I/O connectors including AvBus, 8 DIP switches, 2-push buttons, 8 LEDs, digital thermometer, RS-232 port and JTAG port. The included demo routines sequence the LEDs, read the switches and buttons, sense and display temperature and control the RS-232 interface.

ADS-XLX-SP2-EVL \$149.00 USD resale

ADS-SP2-MB-EVL \$499.00 USD resale - Bundled with MicroBlaze Core License

Xilinx Virtex™-II Evaluation Kit

This kit is available with an XC2V1000 prototyping environment to effectively demonstrate the enhanced benefits of the leading edge Xilinx FPGA solution. Demonstration VHDL code is included with the Kit to exercise standard peripherals on the evaluation board for a quick start to device familiarization.

ADS-XLX-V2-EVL1000 \$400.00 USD resale

Xilinx Virtex™-II High-Speed Evaluation Kit

Engineers use the Xilinx Virtex-II High-Speed Evaluation Kit from Avnet Design Services as a platform to develop and test designs that are targeted to the Xilinx Virtex-II FPGA family. A Xilinx XC2V40 device, capable of implementing over 40,000 system gates, is installed as the target for these designs.

Thorough application notes make use of the hardware features of the evaluation board to demonstrate many of the advanced benefits of the Virtex-II family, including: SelectI/O, Digital Clock Manager (DCM), Xilinx Controlled Impedance Technology (XCITE), SelectRAM™ and Data Encryption Standard (DES) support for design security.

ADS-XLX-XV2-EVL \$149.00 USD resale

Xilinx Virtex™-E Evaluation Kit

This kit is built around a Xilinx XCV100E-6PQ240C. The board included in this kit features a Xilinx SPROM, I/O connectors including AvBus, 8 DIP switches, 2 push-buttons, 8 LEDs, a dual-digit 7-segment LED display, a digital thermometer, an infrared transceiver, a RS-232 port and a JTAG port.

ADS-XLX-VE-EVL \$149.00 USD resale

ADS-VE-MB-EVL \$499.00 USD resale - Bundled with MicroBlaze Core License

Xilinx XPLA3 Evaluation Kit

The Xilinx XPLA3 family of CPLDs feature low operating and stand-by power characteristics and pin-to-pin delays as low as 5ns. The XCR3256XL device found on the XPLA3 Evaluation Kit features 256 macrocells with 6,000 useable gates, system frequencies to 140MHz, in-system programming, 7.5ns pin-to-pin delay and 4 global clocks.

When used with the free Xilinx WebPACK™ software and supplied JTAG cable, the XPLA3 CoolRunner™ Evaluation Kit becomes a complete reprogrammable development and prototyping environment that can help shorten and simplify the design cycle.

ADS-XLX-X3-EVL \$149.00 USD resale

AvBus Adapter

This printed circuit board provides adaptability to both male and female AvBus connectors for interface cable fabrication or vertical height extension. The board is unpopulated.

ADS-AVBUS-ADAPT \$35.00 USD resale

User Prototyping Module

This module provides a user prototype area which interfaces to a single AvBus connector for use with development and evaluation kits. It features an 0.1" grid prototyping area, various surface mount footprints and power rail decoupling capacitors.

ADS-PROTO-DAU \$95.00 USD resale

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